Basic Docker Commands

1. Working Without the sudo Command

Normally, when operating Docker, the `sudo` prefix must be added, as shown below:

```
sudo docker version
```

However, after adding a docker user group, the `sudo` prefix is no longer needed. The way to add the docker user group (run the commands on the host that runs Docker):

```
sudo groupadd docker          # create a docker user group
sudo gpasswd -a $USER docker  # add the current user to the docker user group,
where $USER is automatically resolved to the currently logged-in user
newgrp docker                 # update the docker user group
```

After running the above commands, test with the `docker images` command. If no error is reported, it means you can now use Docker without `sudo`. If an error like the following appears:

```
pi@ubuntu:~$ docker images
WARNING: Error loading config file: /home/pi/.docker/config.json: open
/home/pi/.docker/config.json: permission denied
```

Then run the following commands on the host to resolve it:

```
sudo chown "$USER":"$USER" /home/"$USER"/.docker -R
sudo chmod g+rwx "/home/$USER/.docker" -R
```

2. Basic Docker Commands

1. Pull an Image

⚠ If the image cannot be pulled, you may need to access the internet through a VPN.

```
docker pull
```



2. List Images

```
docker images
```

```
pi@raspberrypi:~ $ docker images
REPOSITORY                    TAG          IMAGE ID       CREATED             SIZE
192.168.2.51:5000/ros-humble  5.1.0        0d9e239e4ecd   About an hour ago   17GB
nginx                         latest       f1d0ee1e24a1   4 weeks ago         198MB
langgenius/dify-web           1.6.0        8410a8e6aaf0   5 weeks ago         533MB
langgenius/dify-api           1.6.0        935c77ebab80   5 weeks ago         1.97GB
redis                         6-alpine     b45d9b799914   5 weeks ago         30.4MB
langgenius/dify-plugin-daemon 0.1.3-local  9a4912493da2   6 weeks ago         1.29GB
postgres                      15-alpine    f490b1e1368a   2 months ago        266MB
langgenius/dify-sandbox       0.2.12       5c2ad4db0b19   3 months ago        636MB
ubuntu/squid                  latest       0164cf4b7a88   4 months ago        244MB
semitechnologies/weaviate     1.19.0       b7fde6d5a98e   2 years ago         51MB
pi@raspberrypi:~ $
```

Parameter explanation:

REPOSITORY — the repository source of the image

TAG — the tag of the image

IMAGE ID — the ID of the image

CREATED — the time when the image was created

SIZE — the size of the image

3. List the Docker Containers That Are Currently Running or Have Been Run

```
docker ps #list running containers
docker ps -a  #list all containers that have ever been run
```

```
pi@raspberrypi:~ $ docker ps
CONTAINER ID   IMAGE                                 COMMAND       CREATED          STATUS          PORTS     NAMES
d5b85c17e6a7   192.168.2.51:5000/ros-humble:5.1.0    "/bin/bash"   11 seconds ago   Up 10 seconds             adoring_bassi
pi@raspberrypi:~ $
```

The figure below shows a list of all the Docker containers that have ever been run.

```
pi@raspberrypi:~ $ docker ps -a
CONTAINER ID   IMAGE                                 COMMAND                  CREATED              STATUS                      PORTS     NAMES
d5b85c17e6a7   192.168.2.51:5000/ros-humble:5.1.0    "/bin/bash"              About a minute ago   Up About a minute                     adoring_bassi
663b22d652f4   192.168.2.51:5000/ros-humble:5.1.0    "/bin/bash /root/run…"   52 minutes ago       Exited (0) 25 minutes ago             thirsty_mclaren
9661ce9c395e   192.168.2.51:5000/ros-humble:5.1.0    "/bin/bash /root/run…"   53 minutes ago       Exited (0) 52 minutes ago             blissful_heisenberg
pi@raspberrypi:~ $
```

In the figure, **Up** indicates a running container, and **Exited** indicates a Docker container that has been stopped.

4. Exit a Container Using the exit Command

```
pi@raspberrypi:~ $ ./run_humble.sh
access control disabled, clients can connect from any host
------------------------------------------------------------
ROS_DOMAIN_ID: 61 | ROS:
my_robot_type: A1 | my_lidar: tmini | my_camera: nuwa
------------------------------------------------------------
root@raspberrypi:/# exit
exit
pi@raspberrypi:~ $
```

5. Restart a Docker Container That Has Been Stopped

```
docker start (container id or container name)   # start a container
docker restart (container id or container name) # restart a container
docker stop (container id or container name)    # stop a container
docker kill (container id or container name)    # force-stop a container
```



6. Delete Containers

```
docker rm container id  # delete the specified container
docker rm -f $(docker ps -a -q)   # delete all containers
docker ps -a -q|xargs docker rm   # delete all containers
```